

Theory of Automata — DFA Construction

(Assignment 2)

Alphabet fixed for all questions (other than Question 48 and 49): $\Sigma = \{a, b\}$

Assignment Details

- **Total Marks:** 100
- **Due Date:** Friday, 12 September 2025, 11:59 PM
- **Submission:**
 - A soft copy (image(s) of the handwritten solution combined into a single PDF) must be uploaded on GCR **before Friday, 12 September 2025, 11:59 PM**.
 - A hard copy must be submitted in Dr. Najmul Hassan's office **by Friday, 12 September 2025, 2:00 PM** (Please put the hard copies of the assignments in the box placed in front of my office. If you cannot find the box, please submit them directly to my office. If I am not available, you may submit them to Dr. Hafeez ur Rehman or Dr. Shams ul Arifeen.).
- **File Naming Convention:** The assignment PDF must be named in the format: Name.RollNo.Semester.Section (e.g., Abubakar_21P9044_5C). Any submission not following this exact format will receive a **20 marks deduction**.

Instructions. For each language below, *construct a deterministic finite automaton (DFA)* over $\Sigma = \{a, b\}$ that recognizes exactly the specified set of strings. Clearly indicate states, start state, accept states, and all transitions.

1. (EVEN–EVEN) Strings with an even number of a 's and an even number of b 's.
2. (EVEN–ODD) Strings with an even number of a 's and an odd number of b 's.
3. (ODD–EVEN) Strings with an odd number of a 's and an even number of b 's.
4. (ODD–ODD) Strings with an odd number of a 's and an odd number of b 's.
5. Strings that begin and end with the same symbol.
6. Strings that begin with a and contain at least two b 's.
7. Strings that end with ab and contain no substring bb .
8. Strings that contain ab as a substring but do *not* contain ba as a substring.

9. Strings that contain both ab and ba as substrings (in any order, possibly overlapping).
10. Strings that start with aa or end with bb (or both).
11. Strings with no occurrence of the substring aaa .
12. Strings with no occurrence of the substring bbb .
13. Strings with no adjacent equal symbols (i.e., strictly alternating symbols).
14. Strings that do not start with b and do not contain bbb anywhere.
15. Strings that avoid aba as a substring.
16. Strings with exactly three a 's.
17. Strings with at most two b 's.
18. Strings with at least three a 's and at most one b .
19. Strings with exactly one occurrence of the substring ab .
20. Strings with exactly two occurrences of the substring ba (occurrences may overlap).
21. Strings in which the number of a 's is a multiple of 3 (no restriction on b 's).
22. Strings in which the number of b 's is congruent to 1 (mod 3).
23. Strings whose length is even.
24. Strings whose length is odd.
25. Strings where the total number of a 's is congruent to 2 (mod 4).
26. Strings that start with b and contain an even number of a 's.
27. Strings that end with a and contain an odd number of b 's.
28. Strings that start and end with a and have a number of b 's divisible by 2.
29. Strings that start with ab and have a total length divisible by 3.
30. Strings that end with ba and have a number of a 's divisible by 3.
31. Strings whose third symbol from the right (if it exists) is a .
32. Strings whose second symbol is b (strings of length < 2 are excluded).
33. Strings that end with a and contain b only in blocks of even length (i.e., b 's appear in pairs).
34. Strings in which every block of consecutive a 's has length at most 2.
35. Strings in which every block of consecutive b 's has length exactly 2.

36. Strings that contain aa but do not contain bb .
37. Strings that contain bb but do not contain aa .
38. Strings that contain aa *exactly once*.
39. Strings that contain bb *exactly once*.
40. Strings that contain *exactly one* of the substrings ab or ba (but not both).
41. Strings in which every a is immediately followed by b (i.e., no a is followed by a or end of string).
42. Strings in which every b is immediately followed by a .
43. Strings where every a is both preceded and followed by b (i.e., all a occur inside bab patterns).
44. Strings where all occurrences of ab are immediately followed by b (i.e., only abb may realize ab).
45. Strings where any b that appears must be part of a block baa (i.e., b only occurs as the first symbol of baa).
46. Strings with an even number of a 's that do not contain bb .
47. Strings with an odd number of b 's that begin with a and end with a .
48. **(Personalized)** Construct a DFA that accepts exactly your name (as a single word) over the appropriate alphabet.
49. Construct a DFA for a simplified “valid email” pattern (state your assumptions clearly).
50. Consider the regular expression

$$((a | b)^* a (a | b)^* b (a | b)^* \cup (a | b)^* b (a | b)^* a (a | b)^*)^*$$

Make DFA for the above expression.

Remember

This assignment is designed to reinforce fundamental concepts in DFA design. Take time to understand each language and justify your automaton design.

Any violation of academic integrity policies will result in severe penalties as outlined in the course syllabus.